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ORIGINAL RESEARCH

Preliminary Results of the 115 kJ Dense Plasma Focus Device IR-MPF-100

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Abstract This work summarizes the design and construction of the first Iranian 115 kJ Mather type plasma focus (PF) machine (IR-MPF-100). This machine consists of a 6.25 cm radius and 22 cm height brass made anode with a 50 mm height insulator which separates the anode and cathode electrodes. Twelve copper made 22 cm height rods play the role of cathode with 10.2 cm radius. Twenty four 6 μF capacitors were used with the maximum charging voltage of 40 kV (maximum energy of 115 kJ) as the capacitor bank and maximum theoretical current around 1.224 MA. The total inductance of the system is 120 nH. By using NE-102 plastic Scintillator, Rogowski coil, current and voltage probes, hard X-ray, current derivative, current and voltage signals of IR-MPF-100 were measured. The primary result of neutron detection by neutron activation counter represents approximately 10^9 neutrons per shot at 65 kJ discharge energy while using deuterium filling gas. Also IR-MPF-100 PF has been tested successfully at 90 kJ by using the argon gas.

Keywords High energy plasma focus · Current derivative signal · Neutron detection · Double pinch structure

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Introduction

Production of short-lived hot and dense plasma column produced by the electromagnetic acceleration of the current sheath is the specific role of the plasma focus (PF) devices [1–4]. This plasma column is hot enough to produce X-rays emission [5] and fusion products like neutrons and ^{3}He, in case of using deuterium filling gas. This machine is also a favorite inexpensive one, for the study of fusion plasmas. The accurate mechanisms of neutron production in PF devices are not clear processes but experimental observations of the pinch column represent that there are three mechanisms which are responsible for the productions of fusion neutrons: thermonuclear process, beam-target and trapped gyrating particles in the pinched plasma. However, the anisotropy studies of the neutrons show that the thermonuclear process has the least contribution and the gyrating particle has the maximum portion at the optimum condition [6].

Every day increments in the applications and new achievements in the field of PF machine [7–14] was the reason for the design and construction of a 100 kJ PF. Besides the formation and acceleration of the current sheath, the discreet characters of dense and hot pinched plasma and consequently the formation of hot spots (plasmoides) in PFs is encouraging, particularly for their use in nuclear fusion of the hydrogen–boron fuel [15, 16]. Usually the final temperature and density of the focused plasma in PFs are 1 keV and $10^{19}/cm^3$ [17] however, the accurate mechanism in the formation of the dense structure in PFs is a challenging issue which still is an open area for more research.

Design and Construction of the Device

One of the goals in the design steps of PF device is the formation of a well-shaped current layer in order to achieve

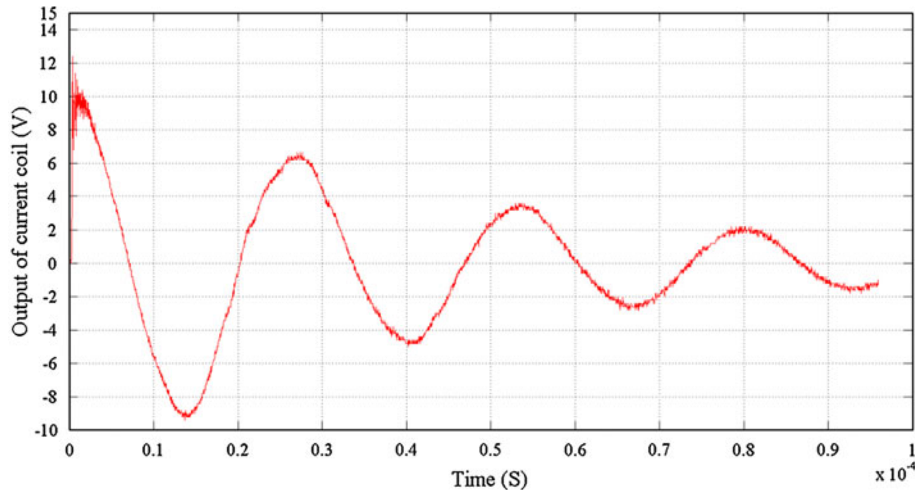

Fig. 1 Discharge current waveform at 21 kV, 30 torr argon

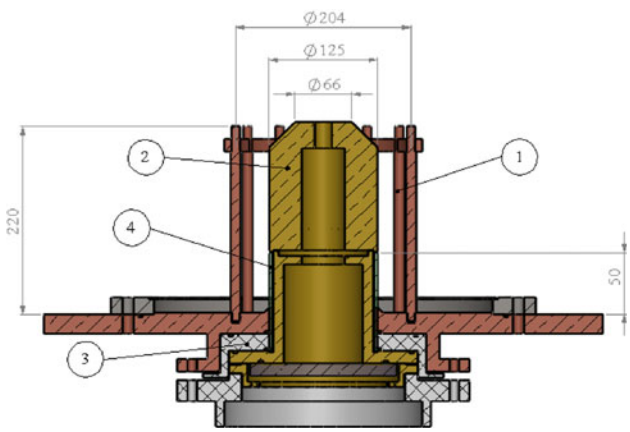

Fig. 2 Schematics of cross sectional view of the (*1*) Cathode rods (*2*) Anode (*3*) PTFE insulator (*4*) Pyrex insulator

Fig. 3 Electrodes configuration of the IR-MPF-100

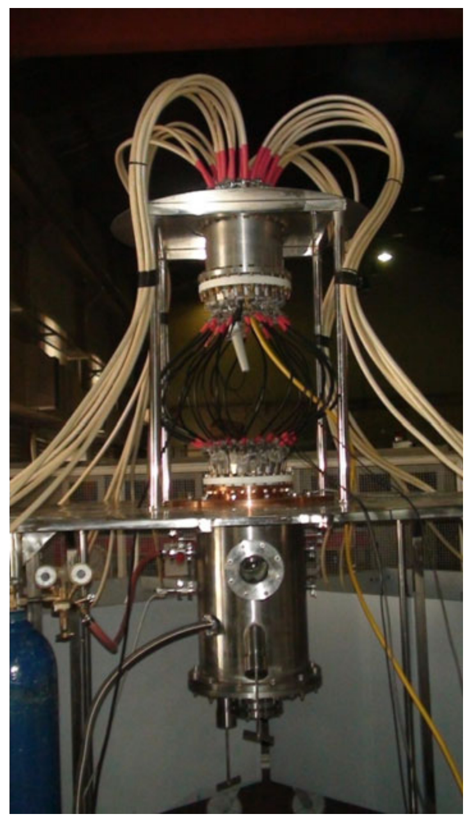

Fig. 4 IR-MPF-100 spark-gape and set up

the top of the anode and finally form a hot and dense plasma column (pinch) another goal is the simultaneous occurrence of the peak current and the pinch effect. For these satisfactions some semi-empirical relations have to be considered in the design of the device [18].

For design of IR-MPF-100 device, we consider a 144 μF capacitor bank (twenty four 6 μF capacitors) with the maximum charging voltage of 40 kV (maximum energy of 115 kJ). Regarding to previous experiments and our current laboratory facilities, the total inductance of the device was estimated to be 120 nH [due to capacitor bank (∼4 nH), spark gap (∼40 nH), coaxial cable (∼60 nH), other connections (∼16 nH)]. By using the following formulas for current, resistance and period of the PF device and discharge current waveform (Fig. 1):

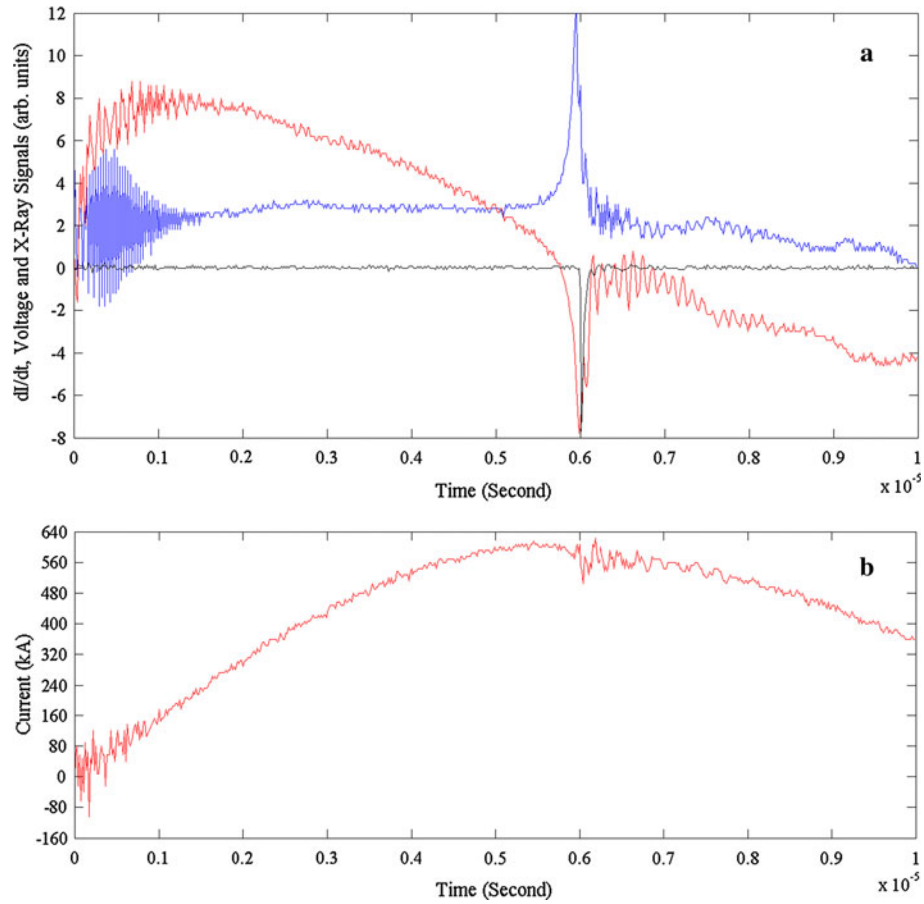

Fig. 5 **a** Hard X-ray, current derivative and voltage signals. **b** Current signal of the IR-MPF-100 at 1.9 torr deuterium and 20 kV

$$I_0 = \pi C_0 V_0 (1 + f)/T \quad (1)$$

$$r_0 = -\frac{2}{\pi} Lnf \sqrt{\frac{L}{C}} \quad (2)$$

In which f is the reversal ratio obtained from the sequential current peaks $I_1, I_2, \ldots, I_n$ with $f = \frac{1}{n}\sum_{i=1}^{n} \frac{I_{n+1}}{I_n} = 0.76$ where "n" is the number of half cycles in the discharge current [19].

$$T = 2\pi\sqrt{LC} \quad (3)$$

The highest Peak current (I_0), resistance and period of the IR-MPF-100 are obtained 1.224 MA, 5 mΩ and 26 µs respectively. On the other hand, the working pressure in the Mather type PF is usually between 0.1 and 10 torr therefore, the considered pressure for this device is 7.7 torr. Taking into consideration that for similar capacitor bank energy of PFs (such as PF-360 [15, 17]), the radius of IR-MPF-100 is taken to be 6.25 cm. According to Ref. [20] the average ratio of cathode to anode radius (c = b/a) approximately is 1.63, therefore this ratio has been selected for IR-MPF-100. Then the cathode radius and speed factor ($S = \frac{I_p/a}{\sqrt{\rho_0}}$, $I_p = 0.7 I_0$) are obtained around 10.2 cm and 50 kA/cm $(\text{torr})^{1/2}$. With respect to Lee's Slug and Snow-Plow Model for the axial and radial phases of current sheath in PF [21], the axial and radial velocity are given according to the following equations:

$$v_{radial} = \frac{\sqrt{\mu(\gamma + 1)}}{4\pi} \frac{f_c}{\sqrt{f_m}} S \quad (4)$$

$$v_{axial} = \sqrt{\frac{\mu \mathrm{LnC}}{4\pi^2 (C^2 - 1)}} \frac{f_c}{\sqrt{f_m}} S \quad (5)$$

On the other hand, $\frac{f_c}{\sqrt{f_m}}$ parameter in Yousefi's paper [20] has reported between 1.3 and 2.2 which was considered to be 1.3 for IR-MPF-100. Now, by using the equations of (4) and (5), axial and radial velocities are obtained 4.037 and 6.071 cm/µs, respectively. The radial time of current sheath (a/v_{radial}) is equal to 1.029 µs then the axial time of current sheath is obtained to be 5.449 µs ($t_{axial} = T/4 - t_{radial}$). So the anode length ($z = v_{axial} t_{axial}$) could be estimated as 22 cm. The space between two electrodes must be in optimum condition to form a suitable final plasma column (pinch). In addition experimental results [22] represent the amount of $\frac{L_{in}}{b-a}$ is usually between 1 and 1.9 therefore we take insulator length (L_{in}) as 5 cm.

Figures 2 and 3 display the cross sectional view and image of the IR-MPF-100 electrodes configuration respectively. Also general set up of device is presented in

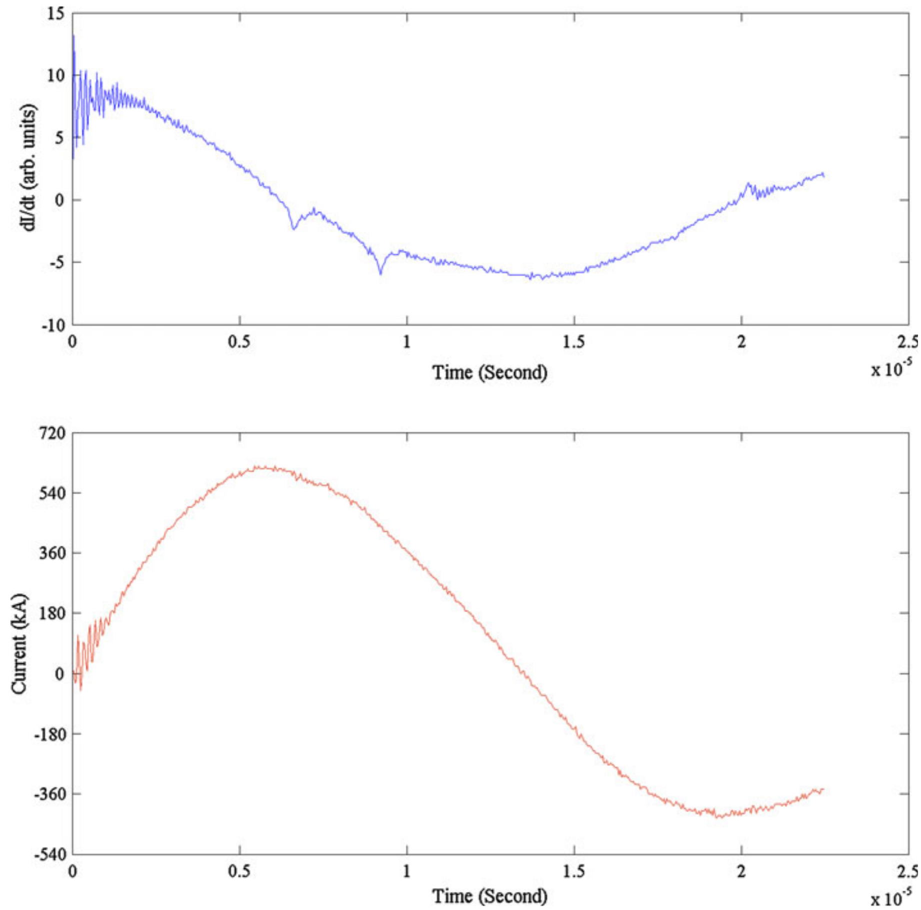

Fig. 6 Double pinch observed in the IR-MPF-100 at 0.3 torr deuterium and 20 kV

Fig. 4. This design has several prominent features such as knife edge which was made in cathode plate near the quartz insulator. The knife edge help form the symmetrical current sheath. The other significant feature is the hollow detachable anode. It is possible to perform various experiments by using the different anode shapes and electron lithography because of this structure.

IR-MPF-100 anode is a 22 cm high and 12.5 cm diameter brass made rode which has a tapered configuration. The brass anode leads to increasing emissions such as neutrons and particularly X-rays. Because the brass anode has low sputtering than copper; therefore, increases neutron yield [4]. Also the brass made up of two components: copper and zinc, where the later has an upper atomic number than former thus, causes to increase X-ray emission. There has been made a central hole in the middle of the anode, in order to prevent the undesirable hard X-ray emissions and the sequential personal dose absorption. Twelve copper rods of the cathode are placed on a 10.2 cm radius circle. The rods of cathode have 12 mm diameter and 22 cm length. Annular ring used at the top of squirrel cage cathode rods which help maintain the distance between rods more accurately. Also this ring suppresses more tensions on the rods and prevents them from bending by the Lorentz electromagnetic force.

Trigatron atmospheric air pressure spark gap switch was used in IR-MPF-100 device. Using one spark gap has the advantage of easy simultaneous discharge of capacitor bank but it has low life time because of high corrosion of the spark gap electrodes and increases the inductance of system.

Results and Discussion

By using NE-102 plastic Scintillator, Rogowski coil, the integrator probe and voltage probe, hard X-ray, current derivative, current signals and voltage signals were measured at the initial pressure of 1.9 torr deuterium gas, which are depicted in Fig. 5. Preliminary experiments of the neutron emission in IR-MPF-100 PF were performed. In order to measure the neutron emission, we utilized neutron activation counter detector which covered by Ag foil. Neutron activation counter detector was placed at 130 cm from top of the anode.

The GM counter was calibrated using a 10 Ci Am–Be source. By using Neutron activation counter neutron emission of IR-MPF-100 was measured nearly 1.5×10^9 n/shot at 29 kJ and 1.9 torr deuterium filling gas. Obviously a double pinch phenomenon was observed

Table 1 Some features of the IR-MPF-100

Parameter	Value
Energy of capacitor bank	115 kJ
Total inductance of the system	120 nH
Capacitor bank	144 μF
Maximum operational voltage	40 kV
Theoretical current	1.224 MA
Speed factor	50 kA/cm (torr)$^{1/2}$
Discharge time period	26 μs
Operating deuterium gas pressure	7.7 torr
Resistance	5 mΩ
Impedance ($\sqrt{\frac{L}{C}}$)	28.87 mΩ
Effective anode length	22 cm
Anode radius	6.25 cm
Cathode radius	10.2 cm
Insulator sleeve length	50 mm

during various shots of the IR-MPF-100 filling Ar gas at 20 kV and 0.3 torr with an approximately time delay of 2.5 μs (Fig. 6). While the current sheath forms on the insulator and moves toward the top of the anode the dynamical resistance increases so that it would be easier for the electrical field between the electrodes to form a new current sheath and move off the insulator while the first current sheath is still in the middle of its way [23, 24] (Table 1).

Conclusion

This paper reports on the design and construction of the IR-MPF-100 PF as the first Iranian 115 kJ PF device. The hollow detachable structure of the anode would make it possible to exchange the anode with different length or structure. And the knife edge of the cathode plate help form the symmetrical current sheath. Also the hollow structure provides a suitable condition for the use of the device in electron lithography experiments. The first experiments performed on the neutron detections represented around 10^9 neutrons per shot using deuterium gas. Regarding to our preliminary experimental results, the neutron yield while using the full energy of capacitor bank (115 kJ) predicted to be more than 10^{10} n/shot and 10^{12} n/shot for D–T fuel.

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